

Commodities

Ingyu Bahng

June 2026

Contents

1	Market Foundation	2
2	Commodity Trading Mechanics	4
2.1	Market Structure & Price	4
2.2	Financialization of Commodities	6
3	Commodity Players	8
3.1	Producers & Consumers	8
3.2	Trading Houses	8
3.3	Investment Banks	9
3.4	Speculators & Investors	9
4	Energy Markets	9
4.1	Crude Oil Fundamentals	9
4.2	Supply Dynamics	10
4.3	Historical Price Shocks	10
4.4	Shale Revolution	11
5	Metal Markets	11
6	Agricultural Markets	11

1 Market Foundation

Core Concept 1.1: Commodity Markets

Commodity markets deal primarily in **physical, tangible assets** that serve as **elementary inputs** for global economic production. Because these raw materials form the base of the manufacturing supply chain, significant price volatility or supply disruptions in these goods inevitably cascade down to consumer-end products. *For instance, while wheat represents the underlying commodity traded in financial markets, commercial bread serves as the final consumer good.*

Modern tradable commodities are systematically classified into three primary sectors.

- **Energy** (Section 4): Primary fuel resources, including crude oil, natural gas, and coal.
- **Metals** (Section 5): Industrial and mineral ores, spanning base, ferrous, and precious varieties.
- **Agriculture** (Section 6): Cultivated assets, comprising grains, oilseeds, soft commodities, and livestock.

Core Concept 1.2: Historical Development of Commodity Trading

Commodity trading represents one of the oldest forms of economic commerce, predating the development of equity and fixed-income markets by several millennia, with historical origins extending back to ancient Mesopotamia.

Early transactions functioned as forward negotiations between parties to exchange a specific quantity of a commodity at a designated future date—a mechanism known today as an **over-the-counter (OTC)** contract. However, persistent disagreements regarding quality standards and delivery terms eventually drove the market toward standardization, giving birth to centralized futures trading. Early global milestones in this institutional evolution include the **Dojima Rice Exchange in Japan** (1697), the **Chicago Board of Trade (CBOT)** (1848) in the United States, and, subsequently the **London Metal Exchange (LME)** (1877) in Europe.

Historically, these centralized marketplaces relied on a physical **open-outcry** system (Figure 1), a framework where floor brokers executed transactions face-to-face within trading pits through synchronized shouting and hand signals.



Figure 1: Chicago Board of Trade open-outcry system termed "the Pit"

In the modern era, however, commodity execution has almost entirely transitioned to electronic platforms. As a result of this widespread digital shift, the LME remains unique as one of the final venues to preserve its legacy open-outcry system termed "the Ring".

Core Concept 1.3: Basic Supply & Demand

Commodity markets are governed by the basic economic principles of **supply and demand** (Section 2). Within this framework, **physical storage** (Figure 2) serves a critical stabilizing function, dampening price volatility and restoring market equilibrium during imbalances, such as an unexpected supply shock.



Figure 2: Japanese oil reserves; in March 2026, Japan announced the release of about 80 million barrels in response to the Middle East conflict in Iran, which has resulted in the disruption of oil flows through the **Strait of Hormuz**

Because production centers are rarely geographically aligned with primary consumption hubs, logistics play a defining role in market dynamics. Over 90% of global commodity trade volume is transported via **maritime freight** (Figure 3), with the remainder distributed across pipelines, rail, trucking, and aviation—meaning that transportation costs represent a significant economic variable that heavily influences both trade routes and ultimate market destinations.



Figure 3: Oil tanker

Accurate analysis of these dynamics, however, is complicated by significant **data acquisition challenges**: sovereign nations report production, consumption, and inventory figures with varying degrees of frequency, scope, and statistical reliability, requiring market analysts to dedicate substantial effort to modeling and estimating these baseline values.

2 Commodity Trading Mechanics

2.1 Market Structure & Price

Core Concept 2.1: Physical Pricing and OTC Structure

Commodity valuation requires the **simultaneous evaluation of multiple physical variables**, where variations in quality, grade, and geographic location heavily influence spot pricing.

In the oil sector, for example, **hundreds of distinct crude grades** are traded globally, meaning the limited selection of standardized futures contracts **cannot always accommodate the precise needs of physical market participants**. Consequently, physical commodity commerce remains predominantly structured as a **bilateral business**, directly connecting buyers and sellers via customized OTC transactions. *This contrasts with exchange-based markets, where standardized contracts enable multilateral participation among a large number of buyers and sellers but offer substantially less flexibility in contract design.*

Core Concept 2.2: Reference Pricing and Benchmarking

To establish reliable baseline values amid this operational complexity, the market relies on specialized **Price Reporting Agencies (PRAs)** such as **S&P Global Platts** or **Argus Media**, which publish independent price assessments across a broad spectrum of spot and derivative contracts. These assessments provide critical transparency into the market, allowing the majority of physical OTC transactions to be executed at an agreed price differential relative to these recognized benchmarks.

Furthermore, because **standardized financial futures do not exist for every unique physical grade**, market participants rely on these benchmarks as pricing proxies to manage asset exposure. In practice, a physical OTC agreement for a niche commodity is structured around a pricing formula pegged directly to a dominant benchmark, adjusted by an agreed-upon **quality or location spread**.

Once the physical transaction is linked to the benchmark, the producer or consumer can execute an equal and opposite position in the financial futures market. This strategy creates a dual-market balancing mechanism: any subsequent loss in physical revenue caused by fluctuating spot prices is offset by a corresponding gain in the futures account, and vice versa. Ultimately, while this mechanism requires participants to **forgo the potential upside of favorable price movements**, it **secures operational certainty by locking in baseline profit margins**.

(a) Producer Hedging Against Brent Price Fall

A producer selling a discounted crude grade can contractually peg their physical sales price to Brent minus \$3.00 per barrel while simultaneously shorting Brent financial futures at \$75.00. If the market declines, the reduced revenue from the physical delivery at \$62.00 is precisely counterbalanced by a \$10.00 gain in the short position, thereby securing the targeted net effective return of \$72.00 per barrel.

(b) Refiner Hedging Against Brent Price Rise

Conversely, a refiner purchasing a discounted crude grade can contractually peg their physical acquisition price to Brent minus \$3.00 per barrel while simultaneously executing a long position in Brent financial futures at \$75.00. If the market rallies, the increased procurement cost from the physical delivery at \$82.00 is precisely counterbalanced by a \$10.00 gain in the long position, thereby securing the targeted net effective purchase cost of \$72.00 per barrel.

Core Concept 2.3: Regional Markets and Arbitrage Dynamics

Crude oil does not trade at a singular, unified global price, operating instead through decentralized regional markets that rely on **localized benchmarks**. Under standard market conditions, the price differentials between these geographic boundaries remain fundamentally constrained; whenever localized imbalances widen these spreads beyond baseline transport costs, market participants exploit the resulting **arbitrage**, shifting physical supply and demand pressures until **price differentials realign**.

Core Concept 2.4: Investment Drivers and Market Financialization

As commodity markets have matured and gained liquidity, they have attracted a growing volume of purely financial participants who do not directly contribute to the physical value chain. This capital influx and participation in commodities markets is structurally motivated by a discrete set of investment objectives.

- **Capital preservation** as a safe haven during periods of broader macroeconomic uncertainty, with tangible assets such as gold serving as archetypal instruments for this purpose
- The use of tangible commodities as **hedgies** to preserve capital during periods of global macroeconomic uncertainty.
- **Portfolio diversification** through exposure to asset classes exhibiting historically low correlation with traditional securities such as equities and fixed income
- Direct **speculative positioning** (*Core Concept 2.7*) to capture profits from the inherent price volatility of the asset class

Like traditional financial assets, commodity spot prices are heavily responsive to shifts in broader market sentiment. This behavioral dynamic has become significantly more pronounced alongside the progressive **financialization of commodity markets** (*Subsection 2.2*), as maturing market liquidity has attracted a growing volume of purely financial participants who do not directly contribute to the physical value chain yet actively trade commodity contracts.

Core Concept 2.5: Price Drivers

The market valuation of primary commodities is dictated by a diverse array of structural, environmental, and institutional drivers.

(a) Weather & Natural Disasters

Environmental conditions and severe weather anomalies exert direct, immediate shocks on regional commodity supply and demand.

For instance, an unseasonal cold wave in the United Kingdom can rapidly escalate residential heating requirements, triggering a sharp increase in regional natural gas prices due to the sudden demand surge.

(b) Geopolitical Events

Because no sovereign nation is entirely self-sufficient, global economic stability depends heavily on an interconnected network of commodity imports and exports. Consequently, any significant disruption to supply or demand dynamics within key trading nations can precipitate adverse cascading effects throughout the global economy.

This is perfectly exemplified by the surge in crude oil prices stemming from the 2026 Iran-US conflict.

(c) Policy & Regulation

Regulatory mandates and institutional policies frequently reshape demand configurations, directly

affecting the profitability of refining and processing firms based on their specific technological and equipment constraints.

A prominent example is the International Maritime Organization (IMO) regulation mandating that all marine vessels utilize fuel with a sulfur content of 0.5% or less, which rapidly altered refinery margins globally by shifting demand toward low-sulfur distillates.

(d) Demand Elasticity

Given that primary commodities function as essential societal **necessities**, aggregate consumer demand tends to be **relatively price-inelastic in the short term**, meaning that broader economic slowdowns typically manifest as **gradual, lagging shifts** rather than immediate disruptions.

There are, however, extreme exceptions to this trend. The outbreak of the **COVID-19 pandemic** rapidly decimated nearly one-third of global fuel demand. This unprecedented collapse in near-term consumption forced the crude oil market into an extreme **super contango**, where immediate physical spot prices traded at a dramatic discount relative to longer-dated futures contracts.

2.2 Financialization of Commodities

Core Concept 2.6: Historical Evolution

While physical supply and demand fundamentals remain foundational drivers of commodity valuation, broader market sentiment also exerts a significant influence on asset prices. This dynamic has intensified over the past few decades due to the progressive **financialization of the global oil market**, characterized by a substantial influx of non-traditional market participants actively trading energy derivatives.

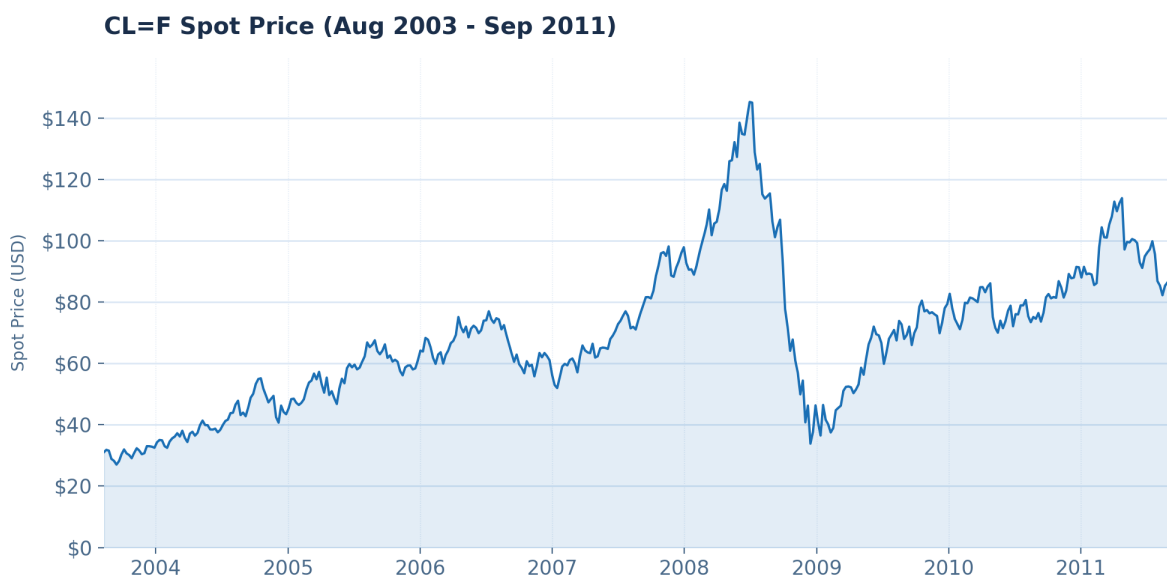


Figure 4: Crude oil future spot from Aug 2003 to Sep 2011

A prominent economic perspective suggests that the dramatic surge in oil prices between 2003 and 2008 (Figure 4) cannot be fully accounted for by physical supply and demand dynamics alone. Instead, this appreciation was heavily facilitated by the rapid financialization of oil futures markets—where trading volumes multiplied by 20 to 30 times and capital allocations expanded from less than \$10 billion to \$450 billion by April 2011—effectively transitioning these physical asset marketplaces into highly financialized arenas.

Core Concept 2.7: Speculation & May 2020 WTI Contract

Speculation involves the acquisition of assets at lower valuation baselines with the intent of selling at higher prices in the future. However, when aggressive speculative capital clashes with severe physical constraints, extreme structural market failures can occur. This vulnerability was vividly exposed during the early stages of the COVID-19 pandemic, when unprecedented global lockdowns triggered an immediate, massive collapse in oil consumption while physical crude production remained highly inelastic, forcing **millions of surplus barrels** into localized distribution networks.



Figure 5: WTI front-month futures prices from January 2020 to December 2020

As illustrated in Figure 5, this supply-demand mismatch culminated in the historic **price collapse of the May 2020 WTI contract**. In the weeks preceding the crash, retail and institutional speculators aggressively channeled capital into crude oil ETFs and passive index products, attempting to capture the market bottom in anticipation of an economic rebound. Crucially, these financial participants overlooked the operational mechanics of the WTI contract, which mandates compulsory physical delivery upon expiration at Cushing, Oklahoma—the primary domestic pipeline crossroads with a working storage capacity of approximately 90 million barrels.

On the eve of expiration, this massive influx of speculative capital encountered an absolute physical bottleneck. Because the global supply glut had filled Cushing's storage facilities to maximum capacity, non-physical market participants holding long positions faced a structural crisis. Lacking the infrastructure to accept physical delivery, these traders were legally required to roll their expiring May contracts into June by liquidating them at any cost. With a massive wall of forced sellers and zero available storage buyers, the market suffered a complete technical breakdown, ultimately driving prices into negative territory for the first time in history.

3 Commodity Players

3.1 Producers & Consumers

Core Concept 3.1: Producers & Consumers

Market participants within the commodity ecosystem are broadly classified into producers and consumers. **Producers** encompass upstream entities such as oil extraction firms and agricultural operations, whereas **consumers** include industrial manufacturing corporations and aviation firms that require physical raw materials for production.

The strategic objectives of these two groups are inherently opposing: producers seek to secure the highest possible selling prices to protect revenues, while consumers aim to lock in the lowest feasible purchase prices to insulate margins. *The aviation sector heavily relies on price locking via futures, as jet fuel procurement frequently accounts for 20% to 50% of total corporate operating costs.*

3.2 Trading Houses

Core Concept 3.2: Role as Intermediaries

Direct transactions between primary producers and end consumers are relatively rare, with specialized **trading houses** typically intermediating these supply chains to manage the underlying logistics and import operations, alleviating the administrative and operational burdens on consuming firms.

Rather than operating strictly as passive intermediaries, however, **modern trading houses frequently integrate vertically** by directly engaging in the extraction, refining, and processing of raw materials. Additionally, they leverage extensive logistical networks to source physical assets from a diversified global supplier base and distribute them efficiently to international end markets.

Core Concept 3.3: Arbitrage Strategies

Trading houses primarily capture profit margins by identifying and exploiting structural **market inefficiencies** through three principal **arbitrage** strategies.

- (a) **Geographical arbitrage** involves purchasing a physical commodity at a lower price in one market and absorbing the freight costs to sell it at a premium in another location.
- (b) **Temporal arbitrage** entails acquiring inventory during periods of localized oversupply or low seasonal demand, managing its storage, and liquidating the asset when prices appreciate in the future.
- (c) **Production arbitrage** leverages processing capabilities, allowing firms to procure lower-cost raw materials and transform them into higher-value refined commodities.

Core Concept 3.4: Major Players

As of June 2026, the global commodity intermediation landscape is highly consolidated, dominated by two principal groups of multinational trading conglomerates.

- **Energy & Metals Merchants:** Glencore (GLEN), Vitol (*private*), Trafigura (*private*), Gunvor (*private*), Mercuria (*private*), Koch (*private*)
- **Agricultural Conglomerates**—collectively known as the “ABCD” participants: Archer-Daniels-Midland (ADM), Bunge (BG), Cargill (*private*), Louis Dreyfus Company (*private*)

3.3 Investment Banks

Core Concept 3.5: Investment Banks

Investment banks fulfill critical financial roles across the commodity value chain by providing essential capital and risk management solutions.

- Structuring project financing for upstream exploration and production firms
- Facilitating equity and debt capital raising
- Offering working capital solutions, such as inventory financing, for midstream and downstream corporations
- Enabling market participants to hedge physical exposure by providing liquidity through commodity-linked financial instruments and derivatives.

Historically, many investment banks actively engaged in physical merchant trading and asset ownership. However, following the **2008 financial crisis**, regulatory reforms and legislative prohibitions compelled most Western institutions to divest from physical operations, though select banks in jurisdictions like Japan and Australia retain these physical capabilities.

3.4 Speculators & Investors

Core Concept 3.6: Speculator vs Investor

Feature	Speculator	Investor
<i>Time Horizon</i>	Shorter	Longer
<i>Trade Frequency</i>	Trade more	Trade less
<i>Risk Appetite</i>	Higher	Lower
<i>Instruments</i>	Futures, Options, Structured Products	ETFs, Mutual Funds

Table 1: Key distinctions between speculators and investors

While both speculators and investors inject necessary capital into the market, they are distinguished by fundamentally different operational profiles as shown in Table 1. Speculators typically maintain short-term transactional horizons with high trading frequencies and exhibit a high risk appetite, primarily utilizing derivative instruments such as futures, options, and structured products. In contrast, investors adopt long-term capital allocation strategies with low transaction frequencies and a more conservative risk tolerance, generally gaining exposure via collective investment vehicles like ETFs and mutual funds.

4 Energy Markets

4.1 Crude Oil Fundamentals

Core Concept 4.1: Brief History of Oil

Core Concept 4.2: Oil Refining and Complexities**4.2 Supply Dynamics****Core Concept 4.3: Seven Sisters & Oil Majors****Core Concept 4.4: Organization of the Petroleum Exporting Countries (OPEC)****Core Concept 4.5: Strategic Petroleum Reserves****4.3 Historical Price Shocks****Core Concept 4.6: Arab Oil Embargo (1973)****Core Concept 4.7: Iranian Revolution (1979)****Core Concept 4.8: Iraq War (1980)****Core Concept 4.9: Price Slump (1980s)****Core Concept 4.10: Price Spike/Fall (1990)****Core Concept 4.11: Asian Economic Crisis (1994)****Core Concept 4.12: Oil Boom (2000s)**

Core Concept 4.13: Great Crash (2008)

4.4 Shale Revolution

Core Concept 4.14: Shale Revolution

5 Metal Markets

6 Agricultural Markets